Composition of Ultra-High-Energy Cosmic Rays

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ABSTRACT

Ultra-High-Energy Cosmic Rays (UHECRs, energy >10^{18} eV) collide with Earth's atmosphere and generate cascades of particles. As the shower develops and loses energy, it reaches a maximum intensity at a point known as X_{max}, a key observable linked to the composition of the primary cosmic ray. By simulating air showers from different nuclear species using the CORSIKA simulation framework, we characterize the statistical behavior of X_{max} distributions and compare how shower development varies across primaries. This work helps constrain the composition of UHECRs and provides accessible reference data for the particle astrophysics community.

COSMIC RAY SHOWERS

How do particles shower?

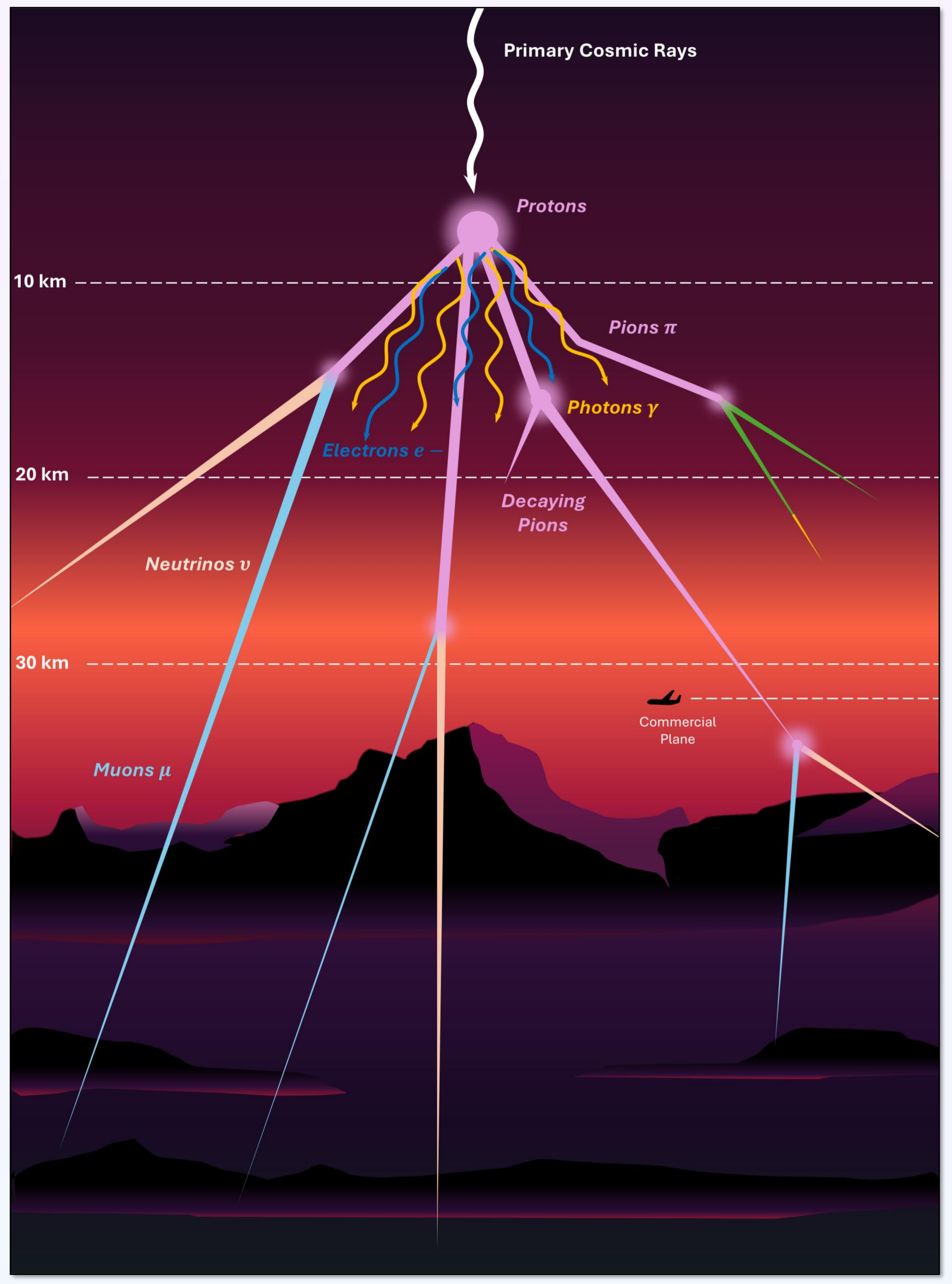

Figure 1. The development of a cosmic ray shower. High-energy primary cosmic rays collide with particles in the upper atmosphere, triggering a cascade of secondary subatomic particles. This visualization tracks the trajectory and decay of pions, muons, and neutrinos from the upper atmosphere (above 30 km) down through the various layers to the ground.

THEORY OF OPERATION

Visualing X_{max}:

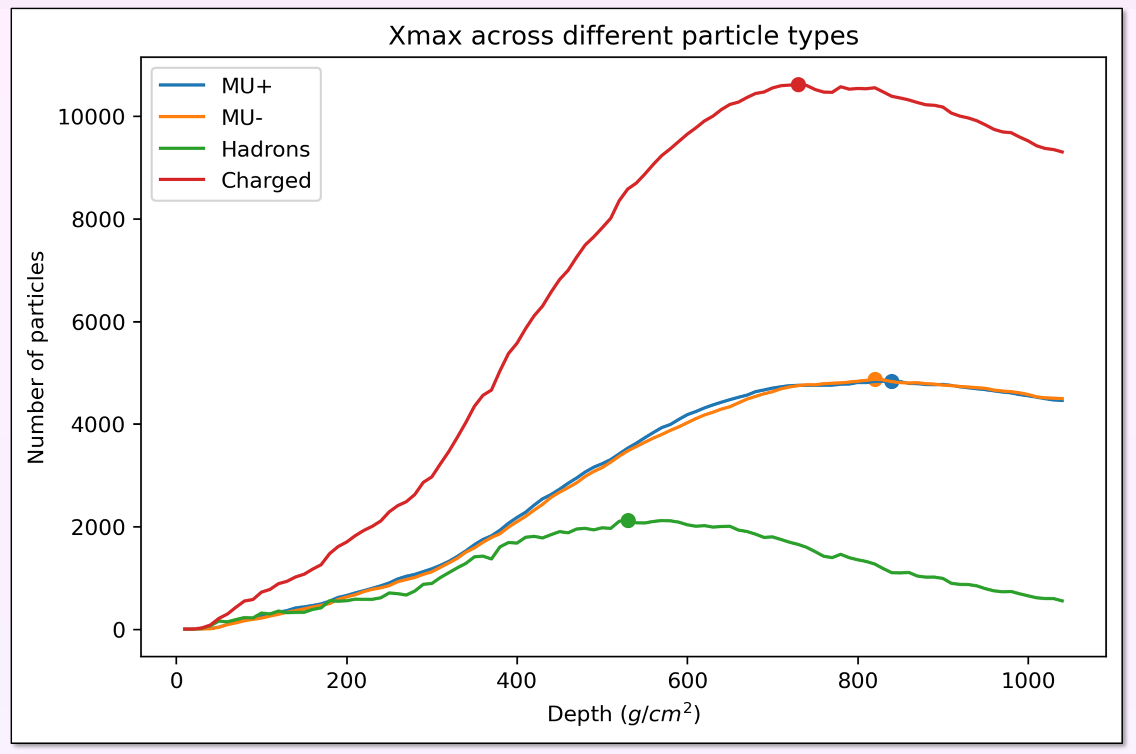

Figure 2. Evolution of particle populations as a function of atmospheric depth, marking the shower maximum X_{max}.

Gaussian fits?

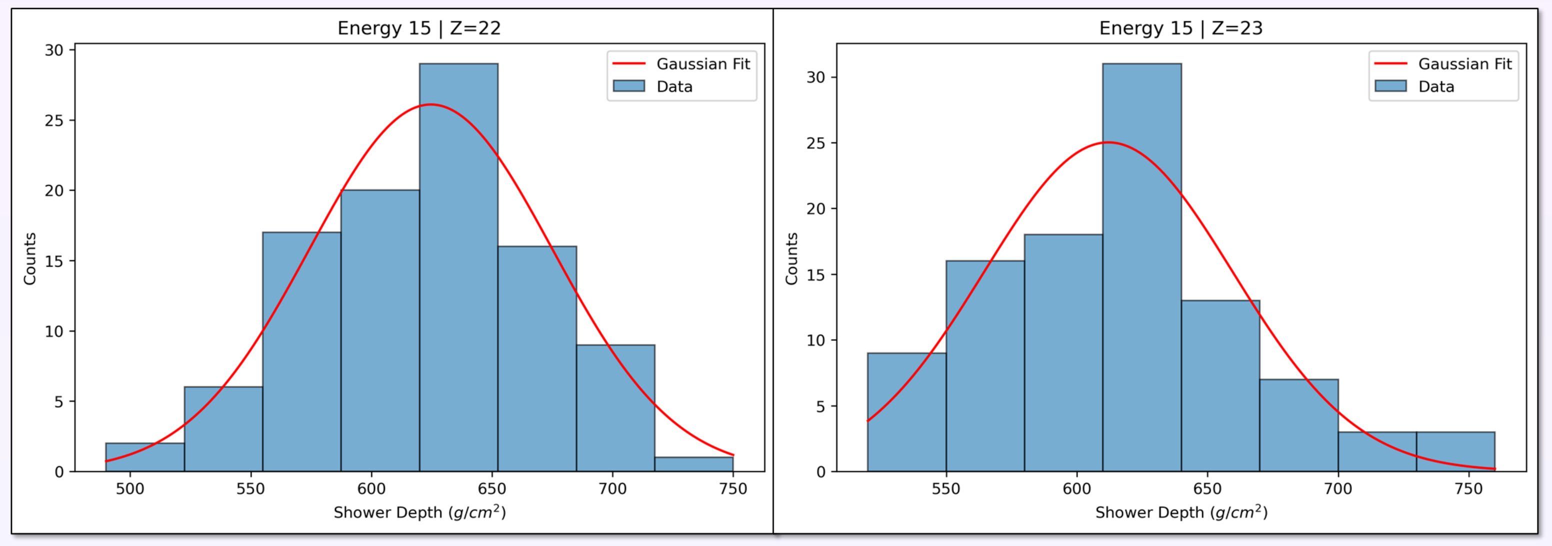

Figure 3. X_{max} distributions and Gaussian fits for titanium (Z = 22) and vanadium (Z = 23) primaries at 10^{15} eV. The overlap indicates that distinguishing these mass groups requires additional observables beyond shower depth.

Across energy levels...

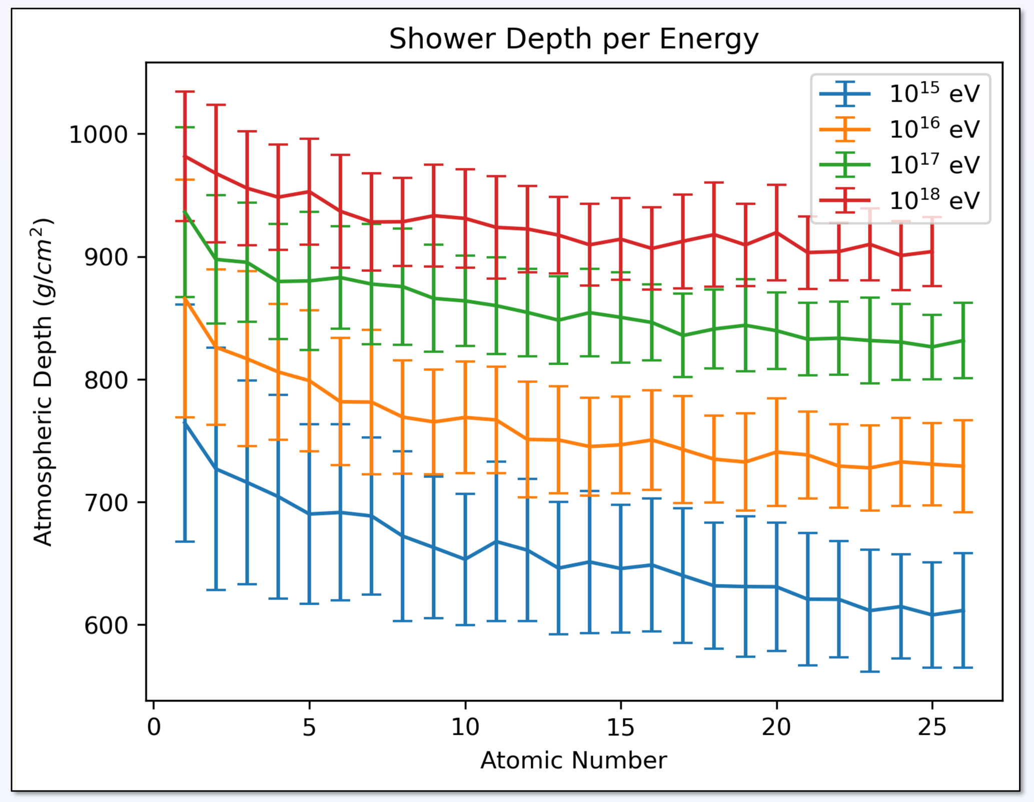

Figure 4. X_{max} scaling with atomic number (Z) and the primary energy (10^{15} – 10^{18} eV)

PARTICLE COMPOSITION

Can we determine particles based on X_{max}?

a)

b)

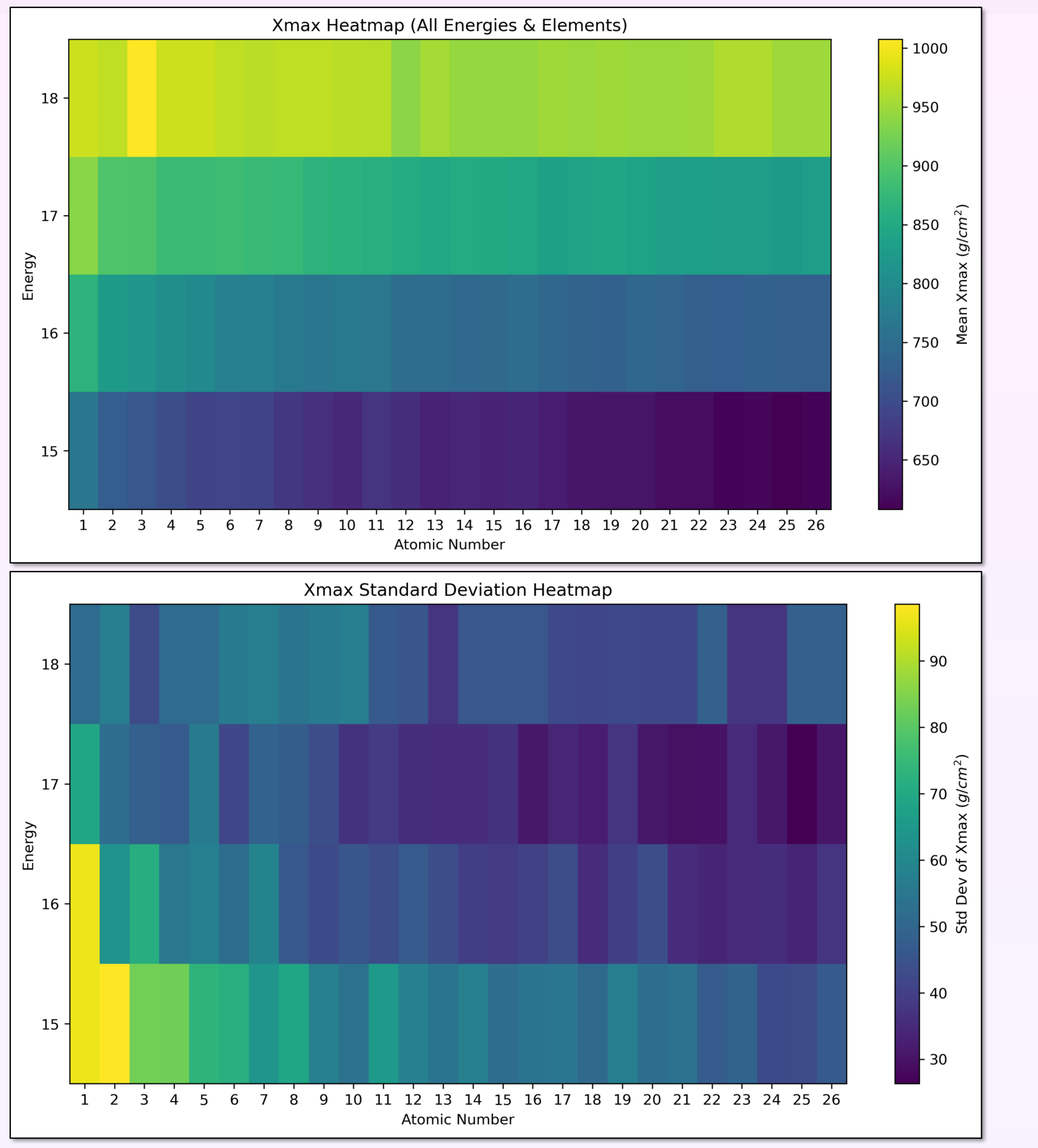

Figure 5. Heatmaps of charged particles showcasing energy, atomic number, and X_{max}. (a) Mean X_{max}: illustrates how different combinations of particle mass and energy produce the same average shower depth, making it difficult to uniquely identify the primary particle. (b) Standard Deviation of X_{max}: Shows that light elements fluctuate significantly more than heavy elements

NEXT STEPS

From these findings, there was no way to determine individual particle composition from X_{max} data. Moving forward, we will look for complementary signals in the particle shower to identify the primary particle, such as muonic content and early Cherenkov light.

ACKNOWLEDGEMENTS

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